



Stability in the frequency domain:
In the root locus approach, when poles go to right half plane, then the system becomes unstable.

However, in the frequency domain, the stability is defined in a different way. It is related to the phase of the system. If the phase lag is 180° degree, the error of the closed loop system may grows exponentially, and

the system may become unstable.

$$e = \overset{\text{reference}}{\hat{R}} - \overset{\text{output}}{\hat{Y}} = \text{instability}$$

There are two quantities that are used to quantify the relative stability of the system in the frequency domain.

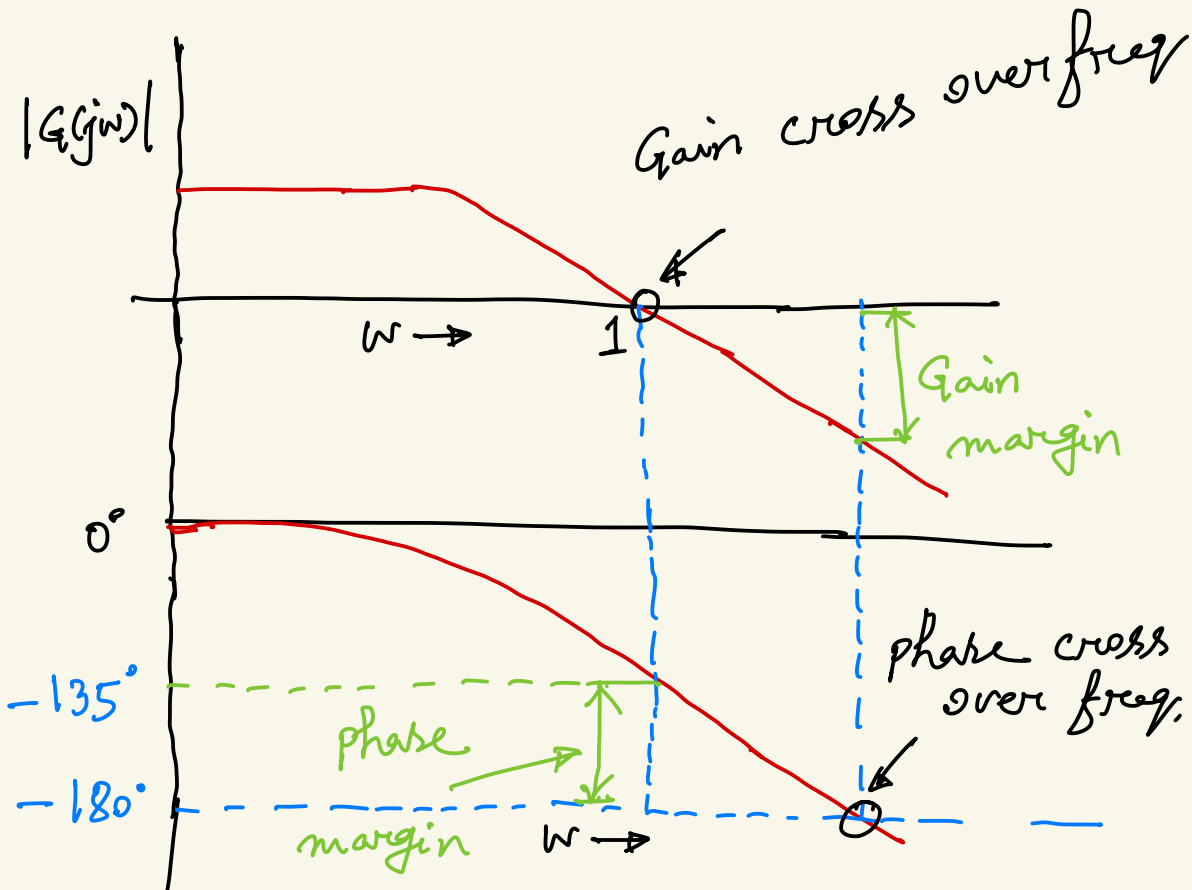
1. Gain Margin
2. Phase Margin

Gain margin: How far away the response is from the magnitude

of 1, when the phase $\phi = -180^\circ$.

phase margin: How far away the response is from -180° phase at gain crossover frequency.

$$|G(j\omega)| = 1 \text{ or } 0 \text{ decibel.}$$



In the above figure,

$$\begin{aligned}\text{phase margin} &= -135 - 180^\circ \\ &= +45^\circ \text{ (positive} \\ &\quad \text{phase} \\ &\quad \text{margin)}\end{aligned}$$

This means that the system can afford 45° phase lag before it becomes unstable. A positive phase margin is required for stability. For any reason, if the phase margin becomes negative, the system will be unstable. At that time, a

compensator can be added to have a positive phase margin, so that the system is stable again.

practice problem:

A unity-feedback control system has the following open-loop transfer function:

$$G(s) = \frac{k \cdot 20(s+5)}{s(s+2)(s+10)}$$

where, $k > 0$, is a variable proportional gain.

1. for $k=1$, derive the exact magnitude and phase expressions

of $G(j\omega)$.

2. Identify the gain-cross over and phase cross-over frequency.